

ENGINEERING MATHEMATICS-I
SUBJECT CODE: NBS-104

COURSE OBJECTIVE: This course covers important mathematical concepts and techniques required in the analysis, design, and optimization of engineering problems. The topics include functions, limits, derivatives, integrals, differential equations, vectors, matrices, and linear algebra.

SYLLABUS:

Unit-I (Matrix Algebra) (8 lectures)

Elementary Transformations, Rank of matrix (Normal form and echelon form), Linear dependence, consistency of linear system of equation and their solutions, Characteristic equation, Eigen value and Eigen vectors, Cayley Hamilton theorem.

Unit –II (Differential Calculus) (8 lectures)

Successive differentiations, Leibnitz's theorem, Rolle's theorem, Lagrange's and Cauchy's mean value theorem with geometrical interpretations, Expansion of functions by Maclaurin's theorem and Taylor's theorem, Homogeneous function Euler's theorem, Maxima and Minima.

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Unit –III (Integral calculus) (8 lectures)

Evaluation of Proper and Improper integrals, Double and Triple integrals, Change of order of integration, Beta and Gamma Function and their properties, Lagrange's duplication formula, Dirichlet integrals and its applications.

Unit-IV (Vector Calculus) (8 lectures)

Parametric representation of vector functions, Gradient, Divergence and Curl of a vector and their Physical interpretations, Important vector identities, Line surface and volume integrals, Statement of Stoke theorem, Gauss Divergence theorem and Green's theorem and their applications.

Unit –V(Ordinary Differential Equations) (8 lectures)

First order Differential equations (linear and nonlinear), Exact Differential equations, Solution of Linear differential equation with constant coefficient of any order, Simultaneous linear Differential Equations.